

Log abundance in numerical dimension one for threefolds in positive characteristic

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Abstract

We prove the numerical-dimension-one case of log abundance for threefolds over algebraically closed fields of characteristic $p > 3$. If (X, B) is a projective log canonical threefold pair with effective rational boundary, and $K_X + B$ is $\mathbb{Q}$ -Cartier, nef, and of numerical dimension one, then $K_X + B$ is semiample. Neither terminality nor $\mathbb{Q}$ -factoriality of X is required.

1 Introduction

The abundance conjecture predicts that a nef log canonical divisor $K_X + B$ on a projective log canonical pair is semiample: some positive Cartier multiple is generated by its global sections. Semiampleness turns the numerical positivity of the divisor into a morphism. We consider threefolds in positive characteristic and the case of numerical dimension one. For a nef $\mathbb{Q}$ -Cartier divisor L on a threefold, this condition means that, for an ample Cartier divisor H ,

$$LH^2 > 0 \quad \text{and} \quad L^2H = 0.$$

Theorem 1.1. *Let k be an algebraically closed field of characteristic $p > 3$. Let (X, B) be a projective log canonical threefold pair over k , where X is normal, B is an effective $\mathbb{Q}$ -divisor, and $K_X + B$ is $\mathbb{Q}$ -Cartier. If $K_X + B$ is nef and has numerical dimension one, then $K_X + B$ is semiample.*

The boundary coefficients are arbitrary rational numbers allowed by log canonicity, and neither terminality nor $\mathbb{Q}$ -factoriality is assumed for the original variety. The theorem proves the numerical-dimension-one case of threefold log abundance in this characteristic range.

Over the complex numbers, Miyaoka proved abundance for minimal threefolds of numerical dimension one [Miy88]; Kawamata completed abundance for minimal threefolds and gave another proof of this case [Kaw92, Section 4]. Keel, Matsuki, and McKernan established log abundance for log canonical threefold pairs [KMM94, KMM04]. In positive characteristic, Keel’s semiampleness criterion [Kee99] and the basepoint-freeness methods of Cascini, Tanaka, and Xu [CTX15] provided foundations. The threefold minimal model program developed through the work of Hacon–Xu, Birkar, and Birkar–Waldron in characteristic greater than five [HX15, Bir16, BW17], the log canonical and relative extensions of Waldron and Hashizume–Nakamura–Tanaka [Wal18, HNT20], and Hacon–Witaszek’s treatment of characteristic five [HW22].

For abundance, Xu–Zhang proved nonvanishing for terminal minimal threefolds in characteristic greater than five [XZ19]. Witaszek obtained klt nonvanishing and abundance in nef dimension at most two in that characteristic range [Wit21]. The nef dimension $n(L)$ is the dimension of the nef

reduction of L . On a threefold over an uncountable algebraically closed field, $n(L) = 3$ means that no curve of L -degree zero passes through a very general point. Zheng Xu established nonvanishing for pseudoeffective log canonical divisors on projective lc threefold pairs in characteristic greater than three, together with abundance in nef dimension at most two [Xu24, Theorems 1.4 and 1.8]; he subsequently proved abundance in numerical dimension two [Xu26, Theorem 1.1]. For the present problem, the remaining case has numerical dimension one and nef dimension three.

The companion paper [OAI26] treats terminal threefolds with nef canonical divisor of numerical dimension one. Its proof isolates a stronger structural result: certain connected nef Cartier divisors on smooth threefolds cannot exist. We state this cycle obstruction in Theorem 2.1, with all its hypotheses. The present paper constructs such a divisor from a log canonical pair with arbitrary rational boundary. The companion’s obstruction is an essential input to the proof.

The boundary and the proof strategy

After an uncountable field extension and a crepant birational change, we may suppose that the underlying variety is terminal and $\mathbb{Q}$ -factorial. Nonvanishing gives a nonzero effective Cartier divisor $N_0 \sim m(K_X + B)$. We assume for contradiction that $n(N_0) = 3$. The main additional issue is to obtain a relation for a power of the canonical bundle near a component of this *log* canonical divisor. The boundary and the exceptional divisors of a resolution must both be controlled there.

Proposition 3.1 supplies that control for the boundary: it proves $BN_0H = 0$ and shows that every boundary component is either contained in $\text{Supp } N_0$ or disjoint from it. If the intersection were positive, complete-intersection curves would have arbitrarily small ratios of N_0 -degree to anticanonical degree. Quantitative bend-and-break [JLR26, Theorem 1.1] would then produce N_0 -trivial rational curves through very general points. This separation argument is formulated independently of the MMP.

Next we adapt the isolation construction of [OAI26, Section 2.3]. Raising the boundary to its log canonical threshold along N_0 gives a coefficient-one component on a dlt model. A partial MMP, trivial for the resulting full log canonical class, makes this component disjoint from the rest of the divisor. We use Xu’s partial-MMP formulation [Xu26, Section 2.3], whose ray argument adapts [KMM94, Lemma 5.1]. Reduced-boundary adjunction and surface abundance [Tan16, Pos23] then yield torsion on a Cartier multiple of that component. We include the companion’s extension argument across finitely many points, since the required torsion is on the full nonreduced Cartier scheme.

Finally, terminality makes the singular locus finite, so the exceptional divisors of a common resolution obey the same support separation as the boundary. A finite separable cover removes the prime-to- p part of the pluricanonical exponent. A connected component of the pulled-back Cartier divisor, divided by the gcd of its multiplicities, has exactly the numerical, torsion, and canonical properties forbidden by Theorem 2.1.

Section 2 states the obstruction and makes the initial reductions. Section 3 proves boundary separation, and Section 4 isolates a component. Section 5 proves torsion on its Cartier thickening. Section 6 constructs the final cycle and completes the proof.

2 The cycle obstruction and the initial reductions

We first state the precise input from the companion paper. We then reduce Theorem 1.1 to constructing the forbidden cycle. Throughout, a variety is integral. Boundaries are effective $\mathbb{Q}$ -divisors, and a pair (X, B) has X normal and $K_X + B$ $\mathbb{Q}$ -Cartier. We use the usual discrepancy

definitions of log canonical (lc), Kawamata log terminal (klt), divisorial log terminal (dlt), and terminal singularities; see [KM98]. Canonical divisors on birational models are chosen compatibly. Restrictions and intersections of $\mathbb{Q}$ -Cartier divisors are computed using Cartier multiples.

2.1 The obstruction supplied by the companion

For a nef divisor L over an uncountable algebraically closed field, its *nef dimension* $n(L)$ is the dimension of the target of its nef reduction. We use only the following characterization in dimension three: $n(L) = 3$ if and only if no curve of L -degree zero passes through a very general point. Here “very general” means outside a countable union of proper closed subsets. These properties and existence of nef reduction are recalled in [Xu26, Definition 2.1 and Theorem 2.2]. Numerical dimension and nef dimension need not agree.

Theorem 2.1 (Cycle obstruction from the companion). *Let k be an uncountable algebraically closed field of characteristic $p > 3$. There do not exist a smooth projective threefold V , a very ample Cartier divisor H_V , and a nonzero effective Cartier divisor $D = \sum_{\alpha} m_{\alpha} D_{\alpha}$ satisfying all the following conditions:*

- (i) *The D_{α} are distinct prime divisors, $m_{\alpha} \in \mathbb{Z}_{>0}$, the support of D is connected and has simple normal crossings, and $\gcd_{\alpha} m_{\alpha} = 1$.*
- (ii) *D is nef, its restriction to every D_{α} is numerically trivial, and*

$$DH_V^2 > 0, \quad D^2 H_V = 0, \quad n(D) = 3.$$

- (iii) *The line bundle $\mathcal{O}_D(D)$ on the possibly nonreduced Cartier scheme D is torsion.*
- (iv) *For some integer $b \geq 0$ and integers b_{α} , there is an isomorphism on a neighborhood of $\text{Supp } D$,*

$$\omega_V^{\otimes p^b} \simeq \mathcal{O}_V \left(\sum_{\alpha} b_{\alpha} D_{\alpha} \right). \quad (1)$$

This is the obstruction proved in [OAI26, Sections 3–6] for the data listed in its Proposition 2.1. The end of Section 2 of that manuscript explicitly retains only V, D, H_V , the torsion order, and the local canonical relation. Under these hypotheses, its Sections 3–5 prove Corollary 5.4, and its Proposition 6.8 gives the contradiction. Thus the obstruction applies to any cycle with the displayed properties. In particular, the construction below may start from a log canonical divisor with boundary.

The companion uses the torsion normal line and the local canonical relation to control differential forms on the completion along D with poles allowed along D . Cartier descent then makes the Picard group of that punctured formal neighborhood p -divisible. The numerical conditions give a nonzero integer-valued degree homomorphism on the same group, which contradicts that divisibility. We use this obstruction as a theorem; the present proof constructs all its input data for a log canonical pair.

The canonical relation also implies an intersection identity that occurs in the companion. Indeed, a rational section giving (1) has global divisor $\sum b_{\alpha} D_{\alpha} + E$, with $\text{Supp } E$ disjoint from $\text{Supp } D$. Since $D|_{D_{\alpha}} \equiv 0$, for every Cartier divisor A on V we have

$$K_V D A = 0. \quad (2)$$

2.2 Numerical observations

The following elementary observation will convert zero intersection numbers into restrictions that are numerically trivial.

Lemma 2.2. *Let T be an integral projective surface, H an ample Cartier divisor, and P a nef $\mathbb{Q}$ -Cartier divisor on T . If $PH = 0$, then $P \equiv 0$. Moreover, a nonzero effective Cartier divisor on T has positive intersection with H .*

Proof. For any integral curve $C \subset T$, a sufficiently large multiple of H has an effective Cartier member E containing C . The effective one-cycle of E has positive coefficient along C . Every component has nonnegative P -degree, whereas $PE = 0$, so $PC = 0$. This proves the first assertion. The second is positivity of the ample degree of each component of the effective one-cycle of a nonzero Cartier divisor. $\square$

In particular, if N is a nonzero effective nef Cartier divisor on a projective threefold and $N^2H = 0$ for an ample Cartier divisor H , then

$$N|_T \equiv 0 \quad \text{for every prime component } T \text{ of } N. \quad (3)$$

Indeed, writing $N = \sum a_T T$, the nonnegative numbers NHT have positive-coefficient sum $N^2H = 0$, and Lemma 2.2 applies.

2.3 Reduction to a terminal underlying variety

We use three results of Xu for projective lc threefolds in characteristic greater than three: nonvanishing for a pseudoeffective log canonical divisor, termination of its flips in the pseudoeffective case, and abundance when the divisor is nef of nef dimension at most two [Xu24, Theorems 1.4, 1.5, and 1.8]. All permit rational boundaries. The relative MMP and crepant $\mathbb{Q}$ -factorial dlt modifications are available in this characteristic range, including characteristic five [HW22, Theorem 1.2], [Xu26, Theorems 2.20 and 2.27]. We also use projective resolution of threefolds, preserving the regular locus, and resolution of divisor supports, preserving their complement on a regular variety; the required forms of these results are recalled in [OAI26, Section 2.1], from [CP08, CP09, CJS13].

Proposition 2.3. *To prove Theorem 1.1, it suffices to rule out the following data: an uncountable algebraically closed field k of characteristic $p > 3$, a terminal $\mathbb{Q}$ -factorial projective threefold X , an lc pair (X, B) , and a nonzero effective Cartier divisor*

$$N_0 \sim m(K_X + B), \quad m \in \mathbb{Z}_{>0}, \quad (4)$$

such that mK_X and mB are Cartier, N_0 is nef, $\nu(N_0) = 1$, and $n(N_0) = 3$.

Proof. First extend the given field to an uncountable algebraically closed field. Integrality and normality persist, and a log resolution with its discrepancy calculation proves that the base-changed pair remains lc. The canonical divisor can be defined by a rational top form on the smooth locus, so the log canonical class also base changes. Nefness persists by ample approximation, and the prescribed intersection numbers are unchanged.

Semiampleness descends under this extension. After making a generated multiple also divisible by the original Cartier index, it is the base change of a line bundle on the original variety. Global sections commute with field extension, and faithful flatness detects surjectivity of its evaluation map. We may therefore work over the enlarged field.

Write $L = K_X + B$ for the original log canonical class. If $n(L) \leq 2$, Xu's abundance theorem applies. Suppose $n(L) = 3$. Take a crepant projective $\mathbb{Q}$ -factorial dlt modification (X_d, B_d) , and resolve X_d projectively. Run a relative canonical MMP of this smooth resolution over X_d . The MMP with scaling applies by [HW22, Theorem 1.2]. Its output cannot be of fibre type over the three-dimensional birational base, and it gives a projective birational morphism

$$g : X_t \longrightarrow X_d$$

with X_t terminal and $\mathbb{Q}$ -factorial and K_{X_t} nef over X_d . Here terminality is preserved by a canonical MMP: discrepancies do not decrease, and a divisor contracted in a negative divisorial step has strictly positive ordinary discrepancy over the output. Relative pseudoeffectivity causes no restriction for this birational morphism; adding a sufficiently large pullback of an ample divisor makes the canonical class big.

The exceptional divisor $F = K_{X_t} - g^*K_{X_d}$ is g -nef. The negativity lemma gives $F \leq 0$. Since X_d is $\mathbb{Q}$ -factorial, the divisor

$$B_t = g^*B_d - F$$

is effective and satisfies $K_{X_t} + B_t = g^*(K_{X_d} + B_d)$. Crepancy makes (X_t, B_t) lc.

The new log canonical class is nef of numerical dimension one. To check the square after pullback, observe first that $L^2T = 0$ for every prime surface T on the original variety: place T in an effective member of a multiple of the ample divisor in the hypothesis, and use nonnegativity of L^2 on all its components. Projection formula proves the vanishing upstairs. Numerical nontriviality persists because a curve downstairs can be dominated by a curve upstairs. Finally, the no-zero-degree-curve property persists over the birational isomorphism locus. Thus the new class still has nef dimension three.

Relabel this pair as (X, B) . By nonvanishing [Xu24, Theorem 1.4], a positive multiple of $K_X + B$ has an effective representative N_0 . It is nonzero because $K_X + B$ is not numerically trivial. Increasing the multiple makes N_0 , mK_X , and mB Cartier, as required. $\square$

From now until the proof of Theorem 1.1, we work with the data of Proposition 2.3. Fix a very ample divisor H on X . Then $N_0H^2 > 0$, $N_0^2H = 0$, and (3) holds. The next step is to establish the same numerical triviality on the boundary components.

3 Separating the boundary from the log canonical cycle

The numerical relation $N_0 \sim m(K_X + B)$ initially controls the log canonical divisor, whereas Theorem 2.1 requires a relation for a power of the canonical bundle. We now show that the boundary contributes only components of the cycle or divisors entirely disjoint from it. The argument uses maximal nef dimension and bend-and-break, and is independent of the subsequent MMP.

Proposition 3.1 (Boundary separation). *Let X be a normal $\mathbb{Q}$ -factorial projective threefold over an uncountable algebraically closed field, and let B be an effective $\mathbb{Q}$ -divisor. Suppose that N is a nonzero effective nef Cartier divisor with $\nu(N) = 1$, $n(N) = 3$, and $N \sim_{\mathbb{Q}} a(K_X + B)$ for some rational $a > 0$. For every ample Cartier divisor H , one has*

$$BNH = 0. \tag{5}$$

Consequently N is numerically trivial on each prime component of B , and every such component is either contained in $\text{Supp } N$ or disjoint from $\text{Supp } N$.

Proof. We may replace H by a very ample multiple. The number BNH is nonnegative. If it were positive, the identity $(K_X + B)NH = N^2H/a = 0$ would give $K_XNH < 0$.

For each sufficiently large integer l , choose positive integers u_l, v_l such that u_lH and $v_l(H + lN)$ are very ample. Their general complete intersection is an integral curve C_l contained in the smooth locus of X . Indeed, normality gives a singular locus of dimension at most one; the first general member meets this locus in finitely many points, and the second avoids them. Bertini's theorem on the smooth open gives the desired integral curve. Its intersections satisfy

$$\frac{NC_l}{-K_X C_l} = \frac{NH^2}{-K_X H^2 - lK_X HN} \rightarrow 0, \quad (6)$$

and $-K_X C_l > 0$ for large l .

The quantitative bend-and-break theorem of Jovinelly, Lehmann, and Riedl [JLR26, Theorem 1.1] applies to an integral curve in the smooth locus of a projective variety over an algebraically closed field, with negative canonical degree, and any nef Cartier test divisor. In dimension three it supplies, through every closed point of C_l , a rational curve Γ satisfying

$$0 \leq N\Gamma \leq 4 \frac{NC_l}{-K_X C_l}.$$

Fix l so large that the right-hand side is strictly less than one. Since N is Cartier, it follows that $N\Gamma = 0$.

For completeness, these complete intersections pass through points to which the very-general-point condition applies. For the two fixed very ample systems, the incidence of a point of X and a pair of members through it is a product of projective bundles over X , hence is irreducible and dominates X . Restricting to the nonempty open set of pairs giving the curves just considered leaves a dense open incidence and a dominant evaluation map. Its image is constructible and contains a nonempty open subset of X . Over the uncountable field, this open contains a closed point outside the countable union excluded by $n(N) = 3$. The curve Γ through that point is a contradiction. This proves (5).

Write $B = \sum b_T T$ with $b_T > 0$. All numbers NHT are nonnegative, so (5) gives $NHT = 0$ for each T . Lemma 2.2 yields $N|_T \equiv 0$. If T is not a component of N , the canonical section of $\mathcal{O}_X(N)$ restricts to a nonzero section on the integral surface T . Were $T \cap \text{Supp } N$ nonempty, its zero scheme would be a nonzero effective Cartier divisor on T and would have positive ample degree. This contradicts $N|_T \equiv 0$, so the two supports are disjoint. $\square$

Apply Proposition 3.1 to N_0 and set

$$P_{\text{out}} = \bigcup_{\substack{T \text{ a prime component of } B \\ T \not\subset \text{Supp } N_0}} T. \quad (7)$$

This is a closed subset disjoint from $\text{Supp } N_0$. Thus $\text{Supp } B \subset \text{Supp } N_0 \cup P_{\text{out}}$, a union of two disjoint closed sets. We retain this separation while isolating one component of N_0 .

4 Isolating a log canonical component

The effective divisor constructed above may have several intersecting components. We now make one of them disjoint from the others, while preserving the pullback of the nef divisor. The construction adapts [OAI26, Section 2.3]: raise the boundary to a log canonical threshold, then lower the coefficient of one component and run an MMP that is trivial for the restored log canonical divisor. The resulting nefness forces the required disjointness.

Proposition 4.1. *Let k be an algebraically closed field of characteristic $p > 3$. Let X be a terminal, $\mathbb{Q}$ -factorial projective threefold, let (X, B) be log canonical with rational boundary, and let*

$$0 \neq N_0 \sim m(K_X + B)$$

be an effective nef Cartier divisor, where m is a positive integer. Assume that N_0 is numerically trivial on each component of its support, and that every component of B is either contained in or disjoint from $\text{Supp } N_0$. Denote by P_{out} the union of the latter components and put

$$U_X = X_{\text{reg}} \setminus (\text{Supp } N_0 \cup P_{\text{out}}).$$

There exist a projective $\mathbb{Q}$ -factorial klt threefold Y birational to X , a rational boundary G_Y , an effective nef $\mathbb{Q}$ -Cartier divisor N_Y , and a prime divisor S_Y with the following properties.

1. *The pair (Y, G_Y) is log canonical, S_Y has coefficient one in G_Y , and $s := \text{mult}_{S_Y} N_Y$ is positive. For rational numbers $\tau > 0$ and $\delta > 0$,*

$$K_Y + G_Y \sim_{\mathbb{Q}} \tau N_Y, \quad K_Y + G_Y - \delta S_Y \text{ is nef.} \quad (8)$$

2. *The divisor N_Y is numerically trivial on every component of its support. Every component of G_Y is either contained in or disjoint from $\text{Supp } N_Y$.*
3. *The divisor S_Y is disjoint from every other component of N_Y and from $\text{Supp}(G_Y - S_Y)$. In particular,*

$$N_Y = sS_Y, \quad G_Y = S_Y, \quad K_Y + S_Y \sim_{\mathbb{Q}} \tau s S_Y \quad (9)$$

on a neighborhood of S_Y , and $S_Y|_{S_Y} \equiv 0$.

4. *The birational correspondence identifies U_X with an open subset of Y . On a smooth projective common resolution $q : V_0 \rightarrow X$, $r : V_0 \rightarrow Y$ that is an isomorphism over this open,*

$$q^* N_0 = r^* N_Y. \quad (10)$$

The same equality holds after passing to any further common resolution.

One may take $\tau = m^{-1} + \lambda$, where $\lambda = \text{lct}_{(X, B)}(N_0) \geq 0$.

Proof. We first find a coefficient-one boundary component lying over $\text{Supp } N_0$. On a log resolution of $(X, B + N_0)$, the log canonical threshold is the minimum of the finitely many ratios

$$\frac{a(E, X, B)}{\text{ord}_E(N_0)} \quad (\text{ord}_E(N_0) > 0),$$

where $a(E, X, B)$ denotes log discrepancy and the divisors considered include the strict transforms of the components of N_0 . Thus λ is finite and rational, possibly zero, and $(X, B + \lambda N_0)$ is log canonical. There is a log canonical place E of this pair for which $\text{ord}_E(N_0) > 0$.

Take a projective crepant $\mathbb{Q}$ -factorial dlt modification

$$\mu : (W, G) \longrightarrow (X, B + \lambda N_0),$$

with every extracted divisor having coefficient one; see [Xu26, Theorem 2.27]. Set

$$N = \mu^* N_0, \quad \tau = m^{-1} + \lambda.$$

Then

$$K_W + G \sim_{\mathbb{Q}} \tau N. \quad (11)$$

The divisor N is nef and numerically trivial on every component of its support: a curve in that support maps to a point or to a curve in $\text{Supp } N_0$, and the assertion follows by the projection formula.

Every component of G is either contained in or disjoint from $\text{Supp } N$. For strict transforms of components of B this follows from the hypothesis, while those coming from λN_0 are contained in the support. An extracted divisor has center in $\text{Supp } N_0 \cup P_{\text{out}}$, since X is klt and the boundary vanishes outside this union. Its irreducible center lies in one of these two disjoint closed sets. In the first case its order on N_0 is positive, and in the second case the divisor is disjoint from $\text{Supp } \mu^* N_0$.

The center on W of the place E lies in $\text{Supp } N$ and, by crepancy and the dlt property, in $[G]$. Choose a component S of $[G]$ containing this center. The preceding separation then implies $S \subset \text{Supp } N$. This argument also covers $\lambda = 0$; it does not require all components of N to occur in G .

The modification μ is an isomorphism outside $\text{Supp } N_0 \cup P_{\text{out}}$. Indeed it has no exceptional divisors there, and a projective small birational morphism to a $\mathbb{Q}$ -factorial variety is an isomorphism. To recall the latter fact, push down a relatively ample Cartier divisor. A Cartier multiple of its pushdown pulls back to the same multiple upstairs, because there are no exceptional divisors. It has degree zero on every contracted curve, contradicting relative ampleness if such a curve exists.

Choose a rational $\epsilon > 0$ sufficiently small that

$$G - \epsilon S \geq 0, \quad \tau N - \epsilon S \geq 0. \quad (12)$$

Apply the partial MMP with lowered boundary $G - \epsilon S$ and restoration divisor ϵS [Xu26, Proposition 2.21, Definition 2.23, and Lemma 2.24]. Its steps are $(K_W + G - \epsilon S)$ -negative and $(K_W + G)$ -trivial. The lowered pair is dlt, and its log canonical divisor is pseudoeffective by (11) and (12). The termination required here is therefore supplied by [Xu24, Theorem 1.5]. The threefold MMP package used in this application, including in characteristic five, is recalled in [OAI26, Section 2.3].

The partial-MMP conclusion is either a Mori fiber contraction for the lowered log divisor, trivial for the restored divisor, or nefness of the restored divisor minus δ times the transform of S for every sufficiently small rational $\delta > 0$. We first track S , the pullbacks of N , and support separation through the steps, and then exclude the fiber-type alternative. Write $W_i \dashrightarrow W_{i+1}$ for a step, with transforms G_i, N_i, S_i . The equality $K_{W_i} + G_i \sim_{\mathbb{Q}} \tau N_i$ persists by pushforward, so its contracted ray R_i satisfies

$$N_i \cdot R_i = 0, \quad S_i \cdot R_i > 0. \quad (13)$$

In particular S_i cannot be the exceptional divisor of a divisorial step: the negativity lemma would give a contracted curve of negative S_i -degree. Hence S survives throughout the MMP.

Let $a : T \rightarrow W_i$ and $b : T \rightarrow W_{i+1}$ be a common resolution of a step. The divisor

$$F = a^* N_i - b^* N_{i+1}$$

is b -exceptional. If a curve on T is contracted by b , its image on W_i is a point or a curve contracted to the common contraction target. Thus F has degree zero on every such curve by (13). The negativity lemma applied to F and $-F$ gives

$$a^* N_i = b^* N_{i+1}. \quad (14)$$

This applies to divisorial contractions and flips. The same argument for the full log divisors gives

$$a^*(K_{W_i} + G_i) = b^*(K_{W_{i+1}} + G_{i+1}).$$

Consequently the restored pair remains log canonical.

Equation (14) also preserves nefness and numerical triviality on the support. For the latter assertion, lift a curve in $\text{Supp } N_{i+1}$ to a curve on T dominating it. Its image on W_i lies in $\text{Supp } N_i$ or is a point: the inverse image of the support of an effective $\mathbb{Q}$ -Cartier divisor is exactly the support of its pullback. The projection formula now gives degree zero. Boundary components contained in $\text{Supp } N_i$ remain contained in the transformed support. If a boundary component is disjoint from that support, its strict transform on T is disjoint from a^*N_i , hence from b^*N_{i+1} . Since that strict transform maps onto its image on W_{i+1} , the transformed boundary component is still disjoint from $\text{Supp } N_{i+1}$.

These steps also preserve the open complement of the support. Indeed every contracted curve has negative degree against the effective divisor $\tau N_i - \epsilon S_i$, so is contained in $\text{Supp } N_i$. A connected positive-dimensional projective fiber contains a curve through each of its closed points. Connectedness of fibers therefore implies that a fiber meeting the complement of $\text{Supp } N_i$ consists of a single point and does not meet the support. Properness and normality show that the contraction is an isomorphism from this complement onto an open subset of its target. For a flip, this open in the target is $\mathbb{Q}$ -factorial, being isomorphic to the corresponding open of W_i . The new small morphism is therefore also an isomorphism over it.

There is no Mori fiber output. The effective representative $\tau N_i - \epsilon S_i$ persists on each model, as does $\tau N_i - t S_i$ for every rational $0 < t \leq \epsilon$. A curve in a fiber through a general point outside this effective support has nonnegative degree, whereas the log divisor driving a Mori fiber contraction has negative degree on every contracted curve. This rules out the fiber-type alternative for the lowered class and for every such smaller perturbation of the restored class.

Let Y be the resulting model. The partial-MMP conclusion gives (8) for some rational $0 < \delta < \epsilon$. Negative MMP steps preserve the $\mathbb{Q}$ -factorial dlt pair with lowered boundary, so $(Y, G_Y - \epsilon S_Y)$ is dlt. In particular its $\mathbb{Q}$ -factorial underlying variety Y is klt: every place of zero log discrepancy for a dlt pair has center in its reduced boundary, and removing that effective $\mathbb{Q}$ -Cartier boundary strictly increases its discrepancy. Here dlt preservation is used for the lowered boundary; for the restored boundary G_Y we use the crepant comparison just proved.

It remains to see why nefness isolates S_Y . If $T \neq S_Y$ is a component of N_Y , numerical triviality of $N_Y|_T$ and (8) imply that $-S_Y|_T$ is nef. On the other hand, $S_Y|_T$ is an effective $\mathbb{Q}$ -Cartier divisor. If it were nonzero, its intersection with an ample Cartier divisor on the integral projective surface T would be positive, contradicting anti-nefness. Therefore $S_Y \cap T = \emptyset$. Each other component of G_Y is either such a component of N_Y , or is disjoint from its support. This proves the asserted separation and the local identities (9). The equality $N_Y|_{S_Y} \equiv 0$ gives $S_Y|_{S_Y} \equiv 0$.

Finally, all models share the stated open U_X . Resolve the graph of the correspondences among all the models, keeping this smooth open unchanged, to obtain V_0 . Composing (14) with $N = \mu^*N_0$ proves (10). Pullback preserves this equality on every further common resolution. $\square$

For adjunction we replace the isolated log canonical pair by a dlt pair. The klt property of Y ensures that all divisors extracted in this last operation belong to the total transform of S_Y .

Corollary 4.2. *With the notation of Proposition 4.1, there is a projective crepant $\mathbb{Q}$ -factorial dlt modification*

$$f : (Y_d, \Sigma) \longrightarrow (Y, S_Y)$$

such that Σ is reduced,

$$\text{Supp } \Sigma = \text{Supp } f^* S_Y, \tag{15}$$

and f is an isomorphism off S_Y . On a neighborhood of Σ ,

$$K_{Y_d} + \Sigma \sim_{\mathbb{Q}} \tau s f^* S_Y. \tag{16}$$

Moreover, $f^ S_Y$ is numerically trivial on each component of Σ , and Y_d shares the open U_X with X and Y .*

Proof. The pair (Y, S_Y) is log canonical since (Y, G_Y) is log canonical and $G_Y \geq S_Y$. Take a crepant dlt modification extracting only log canonical places. Its boundary Σ consists of the strict transform of S_Y and its exceptional divisors, all with coefficient one. Every extracted divisor has center contained in S_Y : outside S_Y the pair is the klt variety Y . Such a divisor has positive order on S_Y , which proves (15). There are no exceptional divisors off S_Y ; since Y is $\mathbb{Q}$ -factorial, the small-morphism argument in the preceding proof shows that f is an isomorphism there.

Crepancy and (9) give (16). Every curve in Σ maps to a point or to a curve in S_Y , where S_Y is numerically trivial. The projection formula proves the final numerical assertion. Under the identification in Proposition 4.1, U_X is disjoint from S_Y and hence is unchanged by f . $\square$

5 Torsion on the full boundary cycle

The reduced dlt boundary Σ of Corollary 4.2 is a surface to which adjunction and surface abundance apply. The prepared-cycle obstruction, however, requires torsion on an effective Cartier divisor, including its multiplicities. We first give the extension argument that passes from the reduced surface to this possibly nonreduced scheme.

Lemma 5.1 ([OAI26, Lemma 2.4]). *Let T be a normal projective threefold over an algebraically closed field of characteristic $p > 0$, and let E be a nonzero effective Cartier divisor on T . Suppose that E is numerically trivial on every integral component of its support. If*

$$\mathcal{O}_T(E)|_{E_{\text{red}} \setminus Z} \simeq \mathcal{O}_{E_{\text{red}} \setminus Z}$$

for a finite set Z of closed points, then $\mathcal{O}_E(E)$ is torsion.

Proof. We reproduce the proof because both the finite exceptional set and the nonreduced divisor will occur in our application. Replace Z by $Z \cap \text{Supp } E$. At $z \in Z$, write $R = \mathcal{O}_{T,z}$, with maximal ideal $\mathfrak{m}$, and let t_z be a local equation of E . Normality gives

$$H_{\mathfrak{m}}^1(R) = 0, \quad \ell_R(H_{\mathfrak{m}}^2(R)) < \infty.$$

Here the second assertion uses the dimension-three hypothesis. Indeed, present $R = A/I$ with A regular local of dimension n and essentially of finite type over the field. Each nonmaximal localization of R is normal of dimension at most two, hence Cohen–Macaulay. Catenarity gives $\dim A_{\mathfrak{q}} - \dim R_{\mathfrak{p}} = n - 3$ for a prime $\mathfrak{q}$ above $\mathfrak{p}$; the Auslander–Buchsbaum formula therefore gives projective dimension $n - 3$ over $A_{\mathfrak{q}}$. Thus $\text{Ext}_A^{n-2}(R, A)$ is supported only at the closed point. It is a finite module, so it has finite length; local duality gives the assertion for $H_{\mathfrak{m}}^2(R)$.

Choose $j \geq 1$ such that t_z^{j-1} annihilates $H_m^2(R)$ at every point of Z . A single j works because Z is finite. For every integer u , the restriction sequences give a commutative diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathcal{O}_T((u-j)E) & \rightarrow & \mathcal{O}_T(uE) & \rightarrow & \mathcal{O}_{jE}(uE) \rightarrow 0 \\ & & \downarrow & & \parallel & & \downarrow \\ 0 & \rightarrow & \mathcal{O}_T((u-1)E) & \rightarrow & \mathcal{O}_T(uE) & \rightarrow & \mathcal{O}_E(uE) \rightarrow 0. \end{array}$$

In local trivializations the left vertical map is multiplication by t_z^{j-1} . Local cohomology with support in the finite set Z is the direct sum of the corresponding local groups. The vanishing of the ambient H_Z^1 makes the connecting maps from the H_Z^1 of the rightmost terms to the ambient H_Z^2 injective. Naturality and our choice of j therefore imply

$$H_Z^1(T, \mathcal{O}_{jE}(uE)) \longrightarrow H_Z^1(T, \mathcal{O}_E(uE)) \quad \text{is zero for every } u \in \mathbb{Z}. \quad (17)$$

The same j works for every twist: locally the invertible sheaves in the diagram are free of rank one, and the annihilating map is still multiplication by t_z^{j-1} .

Now fix this thickening jE . The ideal of $E_{\text{red}} \setminus Z$ in $jE \setminus Z$ is nilpotent. Lift a chosen generator of $\mathcal{O}_T(E)|_{E_{\text{red}} \setminus Z}$ to local generating sections on a finite open cover of $jE \setminus Z$. On overlaps the lifts differ by sections in that nilpotent ideal times the line bundle. For $u = p^e$ sufficiently large, their u th tensor powers agree: in local frames this is the identity $(a+b)^{p^e} = a^{p^e} + b^{p^e}$, with $b^{p^e} = 0$. They consequently glue to a generator

$$s_j \in H^0(jE \setminus Z, \mathcal{O}_{jE}(uE)).$$

Let s be its restriction to $E \setminus Z$. The obstruction to extending s over E is the image of the obstruction to extending s_j over jE . Equation (17), applied to the exact sequences for sections with support in Z , kills that image. Hence s extends to a section of $\mathcal{O}_E(uE)$.

This extended section has no zeros. On each integral projective surface component F of E_{red} , it is nonzero because it is a generator off Z . If its restriction to F vanished anywhere, its zero scheme would be a nonempty effective Cartier divisor on F , and would have positive intersection with an ample divisor. This contradicts the numerical triviality of $\mathcal{O}_F(uE)$. The argument uses only integrality of F , not normality. The section is therefore a generator on the reduction, and then on E , since an element invertible modulo a nilpotent ideal is invertible. Thus $\mathcal{O}_E(uE) \simeq \mathcal{O}_E$. $\square$

We apply this lemma to the reduced dlt model obtained by isolation. Recall that s is the multiplicity of S_Y in N_Y and $\tau > 0$ is the rational number satisfying $K_Y + G_Y \sim_{\mathbb{Q}} \tau N_Y$.

Proposition 5.2. *For the projective crepant dlt modification $f : (Y_d, \Sigma) \rightarrow (Y, S_Y)$ of Corollary 4.2, there is a positive Cartier multiple M of f^*S_Y such that*

$$\text{Supp } M = \Sigma, \quad M|_{\Sigma_i} \equiv 0 \quad \text{for every component } \Sigma_i, \quad \mathcal{O}_M(M) \text{ is torsion.}$$

Proof. The relations furnished by isolation are

$$K_{Y_d} + \Sigma \sim_{\mathbb{Q}} \tau s f^* S_Y \quad \text{on a neighborhood of } \Sigma, \quad f^* S_Y|_{\Sigma_i} \equiv 0. \quad (18)$$

Let $\sigma : \Sigma' \rightarrow \Sigma$ be the finite S_2 -ification. It is unchanged away from a finite set $Z \subset \Sigma$ [Xu26, Proposition 2.14]. Reduced-boundary adjunction gives an effective rational divisor Δ' such that (Σ', Δ') is a projective semi-log-canonical surface pair and

$$K_{\Sigma'} + \Delta' \sim_{\mathbb{Q}} \sigma^*((K_{Y_d} + \Sigma)|_{\Sigma}) \sim_{\mathbb{Q}} \tau s \sigma^*(f^* S_Y|_{\Sigma}). \quad (19)$$

We use adjunction in the form of [Xu26, Lemma 2.19]; see also [OAI26, Section 2.5]. The S_2 -ification is needed because the reduced boundary itself need not satisfy S_2 . Its codimension-one singularities are nodes, and on its normalization the adjunction boundary includes the conductor with coefficient one; the normalized pair is log canonical. These are precisely the semi-log-canonical conditions needed for surface abundance.

The surface Σ' is projective because it is finite over Σ . The last class in (19) is numerically trivial, by (18) and the projection formula for curves. Posva's abundance theorem for projective semi-log-canonical surfaces in positive characteristic therefore makes $K_{\Sigma'} + \Delta'$ semiample [Pos23, Theorem 3.1]. A semiample numerically trivial rational line bundle is $\mathbb{Q}$ -linearly trivial: a globally generated Cartier multiple defines a morphism contracting every curve, hence constant on each connected component. Its line bundle is trivial there.

Since τs is positive and rational, clearing denominators in (19) yields a positive Cartier multiple M of f^*S_Y with $\sigma^*\mathcal{O}_\Sigma(M)$ trivial. In particular,

$$\mathcal{O}_{Y_d}(M)|_{\Sigma \setminus Z} \simeq \mathcal{O}_{\Sigma \setminus Z}.$$

The divisor M is effective, its reduction is Σ , and it is numerically trivial on every component of its support. Applying Lemma 5.1 on the normal projective threefold Y_d proves torsion on the full Cartier scheme M . $\square$

6 Construction of the prepared cycle

We now combine the isolated boundary with the original log canonical section. The boundary supplies a divisor with torsion normal line; the section supplies a canonical relation near its support. A finite cover removes the prime-to- p part of the canonical exponent. This is the preparation of [OAI26, Sections 2.7–2.8], with the support of the original boundary retained throughout the construction.

Proposition 6.1. *Let (X, B, N_0) be the data of Proposition 2.3 in the maximal-nef-dimension case. There exist a smooth projective threefold V , a very ample divisor H_V , and a nonzero effective Cartier divisor*

$$D = \sum_{\alpha} m_{\alpha} D_{\alpha}$$

having all the properties in Theorem 2.1. In particular, its support is connected with simple normal crossings, $\gcd_{\alpha}(m_{\alpha}) = 1$, and $\mathcal{O}_D(D)$ is torsion.

Proof. Equation (3) and Proposition 3.1 give the hypotheses of Proposition 4.1. Choose its isolated model (Y, G_Y, N_Y, S_Y) and the reduced dlt modification $f : (Y_d, \Sigma) \rightarrow (Y, S_Y)$ of Corollary 4.2. Let M be the positive Cartier multiple of f^*S_Y supplied by Proposition 5.2.

The canonical section and its support. Let V_0 be a smooth projective common resolution of X and Y_d , chosen to dominate the intervening models. Write

$$q : V_0 \longrightarrow X, \quad h_0 : V_0 \longrightarrow Y_d, \quad r = f \circ h_0 : V_0 \longrightarrow Y.$$

These models share the open set

$$U_X = X_{\text{reg}} \setminus (\text{Supp } N_0 \cup P_{\text{out}}),$$

where P_{out} is the union of the boundary components disjoint from $\text{Supp } N_0$. We choose the common resolution to be an isomorphism over this open. Projective resolution and resolution of divisor

supports in dimension three permit this choice [CP09, CJS13]. The pullback equalities through the partial MMP give

$$q^*N_0 = r^*N_Y. \quad (20)$$

The integer m was chosen so that $N_0 \sim m(K_X + B)$ and both mK_X and mB are Cartier. It gives a rational section θ of $\omega_{V_0}^{\otimes m}$ with

$$\operatorname{div}(\theta) = q^*N_0 + m(K_{V_0} - q^*K_X) - mq^*B. \quad (21)$$

Indeed, the right side is an integral divisor linearly equivalent to mK_{V_0} . Put $E_0 = \operatorname{Supp} q^*N_0$. Every prime component in (21) is either contained in E_0 or disjoint from it. For q^*B this follows from boundary separation on X . For a q -exceptional divisor, its irreducible center lies in

$$X \setminus U_X = \operatorname{Supp} N_0 \cup P_{\text{out}} \cup \operatorname{Sing} X.$$

A terminal threefold is regular in codimension two, so $\operatorname{Sing} X$ is finite [OAI26, Lemma 2.3]. The center is therefore contained in $\operatorname{Supp} N_0$, or is disjoint from it. Taking inverse images gives the assertion for the exceptional divisor.

Let F_0 be the union of the components of $\operatorname{div}(\theta)$ outside E_0 . Thus

$$Z_0 := E_0 \cup \operatorname{Supp} \operatorname{div}(\theta) = E_0 \sqcup F_0. \quad (22)$$

The disjointness is the point of this description. Components of q^*N_0 can cancel in (21); we retain all of E_0 when choosing the open set preserved by the subsequent resolution.

Removing the prime-to- p exponent. Write $m = p^b m'$ with $b \geq 0$ and $(m', p) = 1$. Choose a nonzero rational generator η of $\omega_{V_0}^{\otimes p^b}$ and write $\theta = u\eta^{m'}$, where $u \in k(V_0)^\times$. Adjoin one root v of $v^{m'} = u$ and normalize V_0 in the resulting field. This is a finite separable cover, possibly of degree smaller than m' . Over $V_0 \setminus \operatorname{Supp} \operatorname{div}(\theta)$, it is an integral component of the torsor of m' th roots of the invertible section θ . This description is independent of the rational generator η . Since m' is invertible in k , that torsor is finite étale.

Resolve the normalization and the total inverse image of Z_0 , preserving the regular étale open over $V_0 \setminus Z_0$. We obtain a smooth projective threefold and a generically finite separable morphism $\pi : V \rightarrow V_0$ for which the total pullback support of N_0 is simple normal crossings. The morphisms used to compare divisors are

$$\begin{array}{ccccc} V & \xrightarrow{\pi} & V_0 & \xrightarrow{q} & X \\ & & \downarrow h_0 & & \\ & & Y_d & \xrightarrow{f} & Y. \end{array}$$

Here q, h_0, f are birational; π is generically finite and separable. Set $\rho = q \circ \pi$ and $h = h_0 \circ \pi$. By (20), $\rho^*N_0 = (f \circ h)^*N_Y$, while the Cartier divisor supplied by Proposition 5.2 pulls back as h^*M .

The rational section $v\eta$ is a section of $\pi^*\omega_{V_0}^{\otimes p^b}$. The determinant of the differential is a nonzero section of the rational line $\omega_V \otimes \pi^*\omega_{V_0}^{-1}$; denote it by J_π . Its nonvanishing at the generic point follows from separability. Consequently

$$\psi = (v\eta)J_\pi^{p^b} \quad \text{is a rational section of } \omega_V^{\otimes p^b}. \quad (23)$$

Both factors are invertible on $\pi^{-1}(V_0 \setminus Z_0)$: the first is a root of an invertible section, and the second is the Jacobian of an étale morphism. Thus the divisor of ψ is supported on $\pi^{-1}(Z_0)$. By (22), each of its prime components lies in $\operatorname{Supp} \rho^*N_0$ or is disjoint from that support.

A *connected primitive Cartier divisor*. Put $M_V = h^*M$ and choose a connected component C of $\text{Supp } M_V$. Since M is a positive multiple of f^*S_Y , and S_Y is disjoint from the other components of N_Y , the set C is also a connected component of $\text{Supp } \rho^*N_0$. Write the part of M_V supported on C as

$$aD, \quad D = \sum_{\alpha} m_{\alpha} D_{\alpha}, \quad a = \gcd_{D_{\alpha} \subset C} \text{mult}_{D_{\alpha}}(M_V). \quad (24)$$

The divisor D is Cartier because V is smooth, and its multiplicities are positive with gcd one. Its support is connected and simple normal crossings.

Every curve in $\text{Supp } M_V$ maps under h to a point or to a curve in Σ . The projection formula and Proposition 5.2 therefore show that M_V is numerically trivial on every component of its support. Near C we have $M_V = aD$, so $D|_{D_{\alpha}} \equiv 0$ for every α . For a curve not contained in $\text{Supp } D$, effectivity gives nonnegative intersection with D . For a curve in its support, the intersection is zero. Thus D is nef, and every very ample divisor H_V satisfies

$$DH_V^2 > 0, \quad D^2H_V = 0. \quad (25)$$

Torsion also survives on the entire Cartier scheme. Choose $t > 0$ with $\mathcal{O}_M(tM) \simeq \mathcal{O}_M$. The pullback divisor M_V is scheme-theoretically $V \times_{Y_d} M$: its local equation is the pullback of a local equation of M , which is a nonzero divisor since V is integral and h is dominant. Pulling back the trivialization gives $\mathcal{O}_{M_V}(tM_V) \simeq \mathcal{O}_{M_V}$; no flatness assumption on h is needed. Restricting along $D \subset M_V$, and using $M_V = aD$ near C , gives

$$\mathcal{O}_D(taD) \simeq \mathcal{O}_D. \quad (26)$$

Hence $\mathcal{O}_D(D)$ is torsion.

The section ψ of (23) has, on a neighborhood W of C , divisor supported on D . Indeed, its other components are disjoint from C and can be removed from that neighborhood. Thus there are integers b_{α} such that

$$\omega_V^{\otimes p^b}|_W \simeq \mathcal{O}_W\left(\sum_{\alpha} b_{\alpha} D_{\alpha}\right). \quad (27)$$

Numerical proportionality and very general points. It remains to show that no D -trivial curve passes through a very general point of V . Put $N_V = \rho^*N_0$. It is a nonzero nef divisor, numerically trivial on every component of its support by the projection formula. Since $\text{Supp } D$ is contained in that support, we have

$$N_V H_V^2 > 0, \quad N_V^2 H_V = N_V D H_V = 0.$$

Together with (25), the Hodge index theorem gives

$$D \equiv cN_V, \quad c = \frac{DH_V^2}{N_V H_V^2} > 0. \quad (28)$$

We verify that this is numerical equivalence on every curve, rather than only on a fixed hyperplane section, following [OAI26, Section 2.8].

Let $\Gamma \subset V$ be any integral curve. For $l \gg 0$, there is an integral surface $T \in |lH_V|$ containing Γ . To see this, twist the ideal of Γ sufficiently far and then multiply its generators by very ample sections. The resulting linear system has no fixed divisorial component and gives a birational map onto its image off Γ . Bertini irreducibility gives an irreducible general member, which is generically reduced on that birational open. As a Cartier divisor on smooth V , it has no embedded

components, and is therefore integral. On a smooth projective resolution $\tilde{T} \rightarrow T$, let d, n, e be the pullbacks of $D|_T, N_V|_T, H_V|_T$, respectively. Their intersections are

$$d^2 = n^2 = dn = 0, \quad e^2 > 0, \quad de = lDH_V^2, \quad ne = lN_VH_V^2.$$

Hence $d - cn$ has square zero and is orthogonal to e . The Hodge index form is negative definite on the orthogonal complement of the positive-square class e , so $d - cn$ is numerically zero. The projective surjection $\tilde{T} \rightarrow T$ supplies an integral curve dominating Γ : take the closure of a closed point of the fiber over its generic point. The projection formula then gives $(D - cN_V) \cdot \Gamma = 0$. This proves (28).

Finally let $Z_i \subset X$ be the countably many proper closed subsets excluded by the nef reduction of N_0 . Remove from V the non-quasi-finite locus of ρ and all the sets $\rho^{-1}(Z_i)$. These are proper closed subsets, since ρ is dominant and generically finite. Their complement contains very general points because k is uncountable. Every curve through any point of this complement maps to a curve on X . If it were D -trivial, then (28) and the projection formula would make its image N_0 -trivial, contradicting maximal nef dimension. All properties in Theorem 2.1 have now been established. $\square$

Proof of Theorem 1.1. Theorem 2.1 contradicts Proposition 6.1. The maximal-nef-dimension case is therefore impossible. By Proposition 2.3, it remains to apply abundance in nef dimension at most two [Xu24, Theorem 1.8]. This gives semiampleness, and faithful descent along the algebraically closed field extension gives it over the original field. $\square$

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